20 2022 — 4g — 2 Marks

2x3
7 days

Department of Computer Engineering
I- B.Tech. (4YDC) EXAMINATION
CO-10504: COMPUTER PROGRAMMING
Mid Term I (April 2022)
Session 2021-2022 (Sem B)

Time: 1 Hrs. **Max. Marks: 20**

NOTE: Answer to the point and make assumptions wherever required and clearly state them.

Q.NO	Question	Marks	CO
Q.1	Choose/Write the most appropriate answer of the following : i) Programs are run in the ____________ memory and store in ____________ memory. ii) 1_even is a valid variable name (True/ False) iii) _________ command used to rename a directory/file in the Linux/Unix environment. iv) Missing terminating is a example of ______ error.	04	CO1

Q.2	Solve the following where
	int a=20,b=10,c=1,d=3;
	float x=1.5,y=5.6,z=8.3;
	i) a / 3 + x * c – b * d <=a
	ii) ++b / 2 – d++ / (int)y * a + c
	iii) a * y + z % 2 / d && !a || b > c
		03	CO1, CO2

Q.3	Draw a flowchart that accepts a year between 1850 and 2150 and prints the day of the week for 1st Jan of that year. Let the day on the 1st Jan 1970 is Thursday.	03	CO1
Q.4	Solve the following: i) $(4052)_6 = (?)_9$ ii) $(EFG.DC)_{19} = (?)_4$ iii) $(3154)_7 = (?)_{11}$	06	CO1

Q.5	What will be the output of following program with proper justification?
	i) int main()
	{
	int x = 2, y = 1, z = 0;
	switch(x)
	{
	case 2:
	x=4; y=x*1;
	case 1:
	x=3; break;
	default:
	x=1; y=2;
	}
	printf("X = %d , Y = %d , Z = %d", x, y, z);
	return 0; }
	ii) int main()
	{
	int a= -1, b=1;
	if(++a&&b++)
	printf("%d%d", a, b);
	else
	printf("a=%d b=%d", a, b);
	return 0; }
		04	CO2

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July · Dec. - 2019 · I

Shri G.S Institute of Technology and Science
Department of Computer Engineering
CO 10504: Computer Programming
Mid Term #1 (2019-2020) Sem A

Time: 60 mins

Max Marks: 25

Q.No.	Question	Marks	CO
1.	Fill in the blanks: (i) 8 Megabytes contain ________ bits. (ii) In base 16 number system the number follows 9A is ________. (iii) Microsoft Word and Vlc Media player are examples of ________ software. (iv) If a program has used pow(2,3) function, than it will be compiled using ________ command.	02	CO1

2.	Evaluate the following Expression:
	int a=4, c=-3, e = 5;
	float b=12.5, f= 4.5, d;
	(i) d= b >10 || b && c < 0 || a > 0
	(ii) e+= ++a /2 – c++ / (int) b * f + b
		04	CO1

3.	Convert the following numbers: (i) $(983.25)_{11} = ( )_7$ (ii) $(121.211)_3 = ( )_9$	04	CO1
4.	There are four types of fruits Apples, Oranges, Bananas and Grapes. Each student can pick up two fruits. There are some conditions which have to be used to pick up the fruits. Draw a flow chart which can take the name of first fruit as an input and print the names of second fruit or fruits that can be picked up. The conditions are: (i) If you pick an apple you can pick banana. (ii) If you pick orange you can pick grapes. (iii) If you pick grapes you can pick banana.	04	CO1
5.	Convert the following if-else program to switch case:	02	CO3

```
#include<stdio.h>
int main()
{
int x;
scanf("%d",&x);
If(x%10==9)
  printf("The unit digit of x is 9");
else if(x%10==8)
  printf("The unit digit of x is 8");
else if(x%10==7)
  printf("The unit digit of x is 7")
else
  printf("The unit digit of x is other than 7, 8 or 9");
return 0;
}
```

6.	Find the output/ Error of the program with proper justification.	04	CO2, CO3

(i)

```
#inlcude<stdio.h>
int main()
{
  int x=10, y= 20;
  if(!(!x)&&x)
    printf("x=%d",x);
  else
    printf("y=%d",y);
return 0;
}
```

(ii)

```
#inlcude<stdio.h>
int main()
{
 int i=3, j=4;
  switch( i | j )
  {
   case 1:
      printf("case 1 Executed");
      break;
   case 3:
      printf("case 3 Executed");
      break;
   case 4:
      printf("case 4 Executed");
      break;
   case 7:
      printf("case 7 Executed");
      break;
   default:
      printf("default Executed");
      break;
  }
 return 0;
}
```

7. Write a program using if-else to accept a coordinate point in XY coordinate system and determine in which quardrant the coordinate point lies. 05 CO3

For Example:

Input : 7 9

Output: The coordinate point (7, 9) lies in the first quardrant.

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Shri G. S. Institute of Technology and Science
Department Of Computer Engineering
CO 1069: Introduction to Computer Programming
(Mid Term #01)

Time: 1 Hr. Max Marks: 25

Note: Attempt all questions.

Q.1 Write the appropriate answer for the following. (3)

i. If sqrt () function is used in the C program, then you have to compile the program by GCC with .... option.

ii. .... is known as forced conversion.

iii. A mistake in program that causes incorrect results is called .... error.

iv. If a=5 & b=3 are defined as integer and c is in float then c=a/b, output is ....

v. Example of ternary operator is ....

vi. Output of the following program is............

```
for(int i=1;i<=5;i++);
    printf("%d", i);
```

Q.2 Draw a flow chart which takes two numbers a and b as input from user and prints the sum of the squares of all the odd numbers between a and b. (4)

OR

Draw a flow chart to find the value of log x:

$\log(x)= (x-1)- (x-1)^2/2 +(x-1)^3/3$

Q.3. Solve the following (3)

i. $(-123.45)_8 = (?)_{12}$

ii. $(101010101110.110001101)_2 = (?)_{16}$

iii. $(AB5.5D7)_{16} = (?)_{10}$

Q.4. Evaluate the following expressions: (3)

int a=10,b=15,c=5,d=25;

float x=2.5, y=1.5;

i. Z = ++b/2-d++/3*a+c

ii. Z = b<=3 * y || a - b % 3&& c++

iii. Z = a % b / c * 10 + 5 - d

10 20+5-25

Q.5 Answer in brief:

i. When is a switch statement better than if statement?

ii. Differentiate between while and do while.

iii. Differentiate between signed and unsigned int.

Q.6 Write a program to check whether the given number is palindrome or not?

Q.7 Write a program to print the following pattern using loops.

```
S S S S
S     S
S     S
S     S
S     S
S S S S
```

Shri G.S. Institute of Technology & Science
Department of Computer Engineering
CO10504: Computer Programming – Sept. 2016
(Mid Term # 01)

Time: 1 Hr. Max. Marks: 20

Note: Attempt all questions.

Q.1 Write the most appropriate answer of the following : [02]

i) The quality of the monitor is measure in ____________.
ii) Programs are run in the ____________ memory.
iii) Pen drive is the example of the ____________ memory.
iv) If any integer A is represented using 32 bits. Then the range of unsigned integer values is ______.
v) C program are converted into machine language with the help of ____________.
vi) ________ is an invalid variable name.
vii) If pow() function is used in the C program, then you have to compile the program by GCC with ______ option.
viii) ________ Command used to create a directory in the Linux/Unix environment.

Q.2 Draw a flowchart to find the given two triangles (represent by three point) are intersect each other. (Handle all cases of the triangle). [04]

Q.3 Answer the following questions: [04]

i) Convert 5c.a3 (in hexadecimal form) to the binary number system.
ii) Convert 123.987 (in decimal form) to the base 7 number system.

Q.4 The following expression has a single pair of parentheses missing that makes it syntactically wrong and it does not compile. Place a pair of parentheses in proper places to correct the expression and evaluate it. [02]
for x = 2, y = 3, z = 5, w = 7:

x * w + y * z = 7

Q.5 Solve the following expression, where [04]

int w = 0, x = 2.5, y = 5, z = 3, r, s = 4, t = 5, u = -3;
double a = 2.36, b = 3.19, c = 3.0, d = 2.91726;

i) (int)(c * y / z + y / z * c)
ii) x - s * t * - c - u
iii) (float)(x + y < z + w && a > b - 17 * x || ! x < 5)
iv) z++ + s % 2 – u + b / d

Q.6 [04]

(i) What will be the value of j for below-mentioned values of i?

```
switch (i) {
case 2: i = i * i;
case 4: i = i * i;
default: i = i * i;
break;
case 16: i = i * i;
}
j = i;
```

For i = 4, j = ______
For i = 16, j = ______
For i = 1, j = ______

(ii) You need to convert the following for/if statements into an equivalent while/switch-case statements:

```
for(i = 2; i <= sqrt(n); i += 3)
{
    if ( i % 2 == 0)
    {
        i = i + 4 ;
    }
    else
    {
        i = i + 1;
    }
}
printf("%d ", i);
```

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Shri G.S. Institute of Technology & Science
Department of Computer Engineering
CO10504: Computer Programming – Sept. 2017
(Mid Term # 01)

Time: 1 Hr. Max. Marks:

Note: Attempt all questions.

Q.1 Write the most appropriate answer of the following :

i) Register are also knows as ________.
ii) Programs are run in the ________ memory and store in ________ memory.
iii) CUP speed is measure in ________.
iv) If any unsigned integer 'a' is represented using 61 bits. Then the range of unsigned integer values are ________.
v) Pen drives are the example of ________ memory.
vi) _123ABc is an example of ________ variable name and 123_ABC is an example of ________ variable.
vii) ________ command used to rename a directory/file in the Linux/Unix environment.
viii) Save and Quit from the VI editor in Linux by ________ command.

Q.2 Draw a flowchart to find the given number is a prime numbers or not a prime Number.

Q.3 What is the radix/base of the numbers if the solutions to the quadratic equation $x^2 - 10x + 13 = 0$ are $x=5$ and $x=8$.

OR

Consider a quadratic equation $x^2 - 13x + 36 = 0$ with coefficients in base b. The solutions of this equation in the same base b are $x=5$ and $x=6$. Then find the value of base b.

Q.4 What are previous and next number of 7777 in octal (base 8) system?

Q.5 Solve the following expression, where

```
int w = 0, x = 2, y = 15, z = -2, s = 4, t = 5, u = -3;
double a = 2.34, b = 1.23, c = 7.0, d = 2.22;
```

i) `(int)c * y / z + y / (float)z * c)`
ii) `x - s * t / c - u + d++`
iii) `x % y < z * w && a-- > c - 17 * x || ! x < 5 && !w`
iv) `w + z % 2 - u * b + x - 1`

Q.6 Write a program to find the given number is divisible by 2, 3, 5 or 7. If number is divisible by 2, then output is "Number is multiply of 2", If number is divisible by 3, then output is "Number is multiply of 3.", If number is divisible by 5, then output is "Number is multiply of 5.", If number is divisible by 7, then output is "Number is multiply of 7.", If number is divisible by 3 and 7, then output is "Number is multiply of 3. Number is multiply of 7.", If number is divisible by 3, 5 and 7, then output is "Number is multiply of 3. Number is multiply of 5. Number is multiply of 7." ....

a. Write the above program using if-else/if-else-if statements.
b. Write the above program using switch/case statements.

Q.7 Consider a positive integer n of type int. The next higher permutation of n is the smallest integer greater than n which is formed by permuting the digits of n. For example, the next higher permutation for n=1209861 is 1210689, the next higher permutation for n=1421731 is 1423117. Note that next higher permutation may not exist for every number. Write a program to find the next higher permutation of the given number. [05]

142

1421731

++ -- !
* / %
&&
||
right to left

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Department of Computer Engineering
CO10504: Computer Programming – Jan. 2018
(Mid Term # 01)

Time: 1 Hr. Max. Marks: 25

Note: Attempt all questions.

Q. 1. What is a syntax error? Give an example of one in English language and one in C programming language. [01]

Q. 2. Why the few keys on the keyboard are duplicate? Also enlist them. [01]

Q. 3. Identify the data type (as used in declaration statements) and the printf() format specifier for each of the following constants: [02]

Constant	Type	Specifier
a. 12		
b. 0X3		
c. 'C'		
d. 2.34E07		
e. '\040'		
f. 7.0		
g. 6L		
h. 6.0f		

Q. 4. Correct this silly program. (The / in C means division.) [02]

```
void main(int)   / this program is used to count the cow /
{
        cows, legs integer;
        printf("How many cow legs did you count?\n);
        scanf("%c", legs);
        legs / 4 = cows;
        printf("That implies there are %f cows.\n, cows")
}
```

Q. 5. How you measure the following computer hardware? [02]
Secondary Memory, Processor speed, Monitor quality, Printer quality

Q. 6. If $(10010)_2$ binders bind $(384)_{16}$ books in $(12)_8$ days. How many binders will be required to bind $(550)_{11}$ books in $(12)_{10}$ days? [02]

Q. 7. What are previous and next number of $(ABCDD)_{14}$ in base 14 system? [02]

Q. 8. Draw a flowchart to sums all the even numbers between 1 and 200 inclusive and then displays the sum. [03]

Q. 9. Consider a quadratic equation $x^2 - 17x + 70 = 0$ with coefficients in base b. The solutions of this equation in the same base b are x=7 and x=10. Then find the value of base b. [03]

Q. 10. Solve the following expression, where [03]
int w = 1, x = 10, y = 15, z = -20, s = 44, t = 25, u = -13;
double a = 22.34, b = 13.23, c = 17.0, d = 32.22;

i) `y % x + c * w / z + y / (float)z - c`
ii) `x - s * t / c – u + d++`
iii) `s % y < z / t || a > c - 17 * ++x && !x < 5 && !w`

Q. 11. Write a C program to input electricity unit charges and calculate total electricity bill according to the given condition: [04]
For first 50 units Rs. 0.50/unit
For next 100 units Rs. 0.75/unit
For next 100 units Rs. 1.20/unit
For unit above 250 Ks. 1.50/unit
An additional surcharge of 20% is added to the bill

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AB-21080

Shri G.S. Institute of Technology & Science
Department of Computer Engineering
CO10504: Computer Programming – Oct. 2018
(Mid Term # 01)

Time: 1 Hr. Max. Marks: 2[illegible]

Note: Attempt all questions.

Q.1 Write the most appropriate answer of the following : [10]

i) The fastest and smallest memory unit in the computer system is ________.

ii) Generally in the computer system ________ memory is more than ________ memory.

iii) Linux is a ________ software and VI is a ________ software.

iv) If integer 'a' is represented using 128 bits. Then the range of unsigned integer is ________ and ranged of signed integer is ________.

v) Pen drives are the example of ________ memory.

vi) 123.ABc is an example of ________ variable name and 123_ABc is an example of ________ variable.

vii) ________ Command used to delete a directory/file in the Linux environment.

viii) C program are execute by________ and compile by ________.

Q.2 Draw a flowchart to find the given number is perfect power of a three or not. [03]

Q.3 Solve the following: [02]

$$(123.456)_{11} = (\ ?\ )_{15}$$

Q.4 What are previous and next number of FFFF in hexadecimal (base 16) system? [02]

Q.5 Solve the following expression, where [04]

int w =2, x = 3, y = 4, z = -5, s = 5, t = 7, u = -3;
double a = 0.34, b = 12.23, c = 87.0, d = 0.22;

i) `y * w / t + (-1) * s / t * c - u`
ii) `x - s / t * y – u + s++ * a`
iii) `y % w - 1 < u / y && --w > b - 17 * d || ! x < 5 && !w`
iv) `w + z % 2 + c / b / x – 1`

Q.6 Write a program to find the given two circles are interest or not. Yours program should read the two circles from the keyboard and should print 0, if the two circles do not intersect, and it print value 1 if they intersect at exactly one point (i.e. they touch each other) and print 2 if they intersect at 2 different points. Yours program also print the point of intersect, if applicable. One circle is represent by the center of the circle (x,y) and its radius r. [05]

Q.7 Write a program for pascal triangle. Pascal's triangle is a triangular array of the binomial coefficients. Yours program should read the number of rows from the keyboard and print the pascal triangle. For the 8 row, pascal triangle is given by: [05]


```
              1
            1   1
          1   2   1
        1   3   3   1
      1   4   6   4   1
    1   5  10  10   5   1
  1   6  15  20  15   6   1
1   7  21  35  35  21   7   1
```

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Shri G.S. Institute of Technology & Science
Department of Computer Engineering
CO10504: Computer Programming
Mid Term # 01 (Feb 2019)

Time: 1 Hr. Max. Marks: 25

Note: Attempt all questions. This Mid-Test is **OPEN BOOK** test. Only printed materials are allowed.

Q. 1. Choose/Write the most appropriate answer of the following : [05]

i) Pen drive is an example of ______________ memory.

ii) Now a day, size of the secondary memory is measure in ______________.

iii) VI editor is example of ______________ software.

iv) Programs are save in primary memory and run in the secondary memory (True / False)

v) ______________ are the memory of the CPU.

vi) If any integer A is represented using 19 bits in some machine, then the range of unsigned integer values is ______________ to ______________.

vii) $(1234)_{10} + (110011)_2 = (\_\_\_\_\_\_\_\_\_\_)_{10}$.

viii) _123_sum is a valid variable name (True/ False)

ix) Output of the statement printf("%d", if(5 < 5)); is ________.

x) Full form of GCC is______________.

Q. 2. Draw a flowchart to find angle between hour and minute hands of a clock at a given time. [03]

Q. 3. Solve the following where [03]

int a=20,b=15,c=10,d=5;

float x=1.5,y=5.6,z=9.3;

i) a / 3 + x * c – b * d

ii) !!b / 2 – d++ / (int) y * a + c

iii) 2.8 * y + z / 2 / d * a

Q. 4. What is the difference between creating constant using *#define macro* and *const* keywords? **[01]**

Q. 5. What are the benefits of designing a flowchart for solving a problem? **[01]**

Q. 6. What is a variable? Explain the ways to declare scope of a variable. **[01]**

Q. 7. What is meant by control statements? What are the different types of control statements in C languages? **[01]**

Q. 8. Explain the steps involved in solving a problem using computer. **[01]**

Q. 9. Compare if-else-if and switch-case statements with giving examples for their relevant use. **[01]**

Q. 10. Working of computer system is based on binary number system, then why different types of number systems available? **[01]**

Q. 11. What are the purpose of different types of brackets supported by C languages? **[01]**

Q. 12. How run-time error are different from compile-time errors? **[01]**

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